

EEE3096S-L5-Quiz

“Selecting an Operating System for an Embedded System”

Answers

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Selecting OS for an Embedded System Q1

... answer? ...

Task Scheduling is an important factor in selecting an operating system for a real-time embedded system. What is the main reason making Task Scheduling so important for RT systems? Select the best option below.

- ☐ Task scheduling allows for simpler system design, making it easier to spot cases where the real-time deadlines could be missed.
- ☐ Task scheduling makes it impossible for real-time deadlines to be missed, as this method allows any real-time deadline processing event to supersede all other operations.
- ☒ Task scheduling enables deterministic and predictable system behavior, aiding in the analysis and verification of real-time requirements.
- ☐ Task scheduling allows for more efficient use of memory resources.

Selecting OS for an Embedded System Q2

If you are building a prototype that needs to be done quickly, which one of these aspects would you put as highest priority in selecting an operating system? (provide a short motivation in the text section below... there isn't necessarily a perfect choice)

Choose

Speed of Interrupt Handling

Long-term maintenance

Support for paged memory (i.e. memory pages that can be swapped between RAM and hard-drive)

Availability of development tools and library

Power management (e.g. CPU frequency scaling, sleep modes)

Do not submit passwords through Google Forms

obviously...

and motivation...

Selecting OS for an Embedded System Q2

Motivation for the option you chose above.

Your answer

It needs to be developed quickly, so have tools and libraries available will help speed up the process of developing the system. There aren't really any other options in the list that are going to have as significant influence on the development of the system (the next most likely would be interrupt speed, or paged memory – but having usable tools is essentially more important than whether or not paged memory can be used).

end of quiz solution

Any other views or reasons for
having selected other answers?